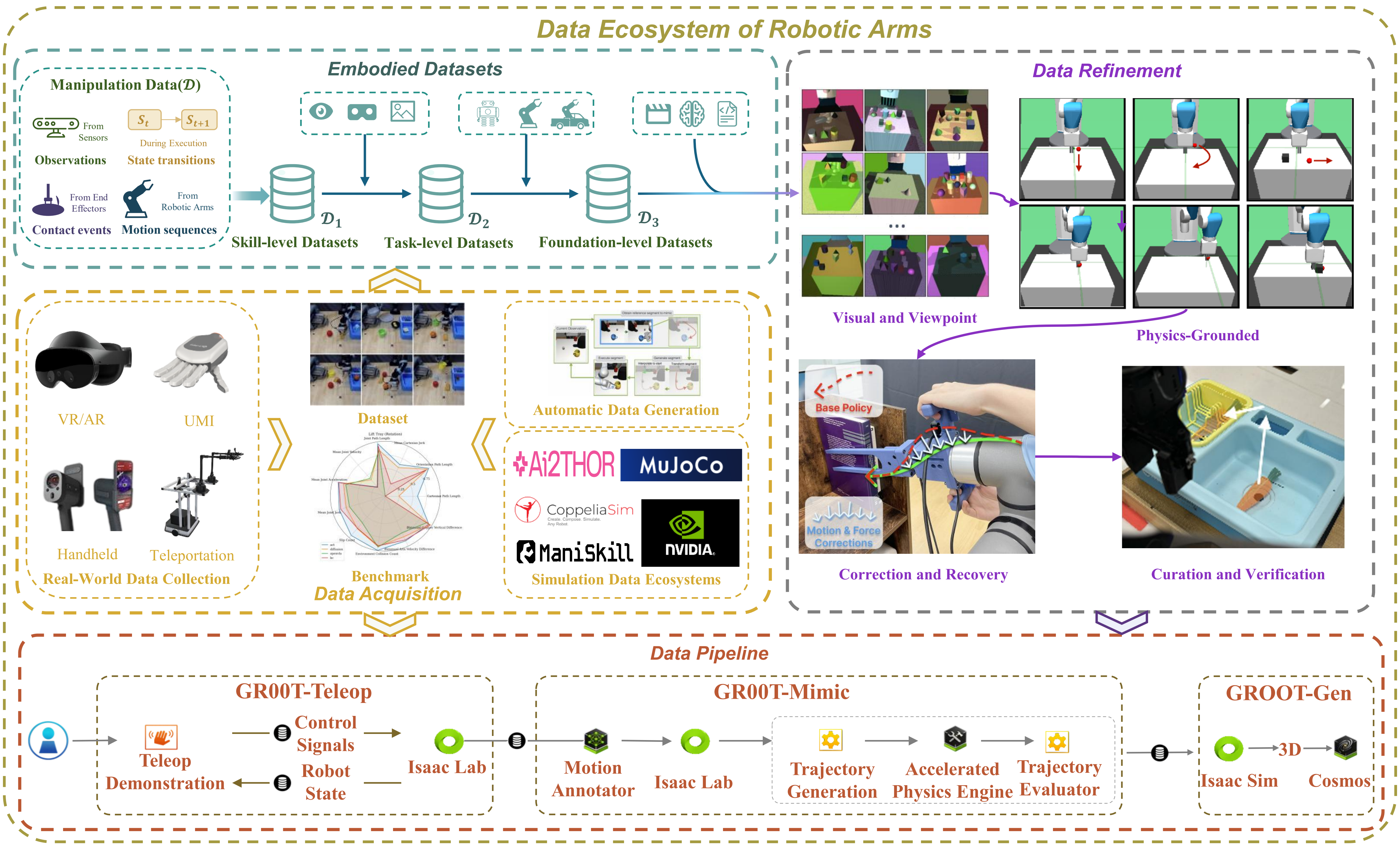

Data Ecosystem of Robotic Arms
Embodied Datasets
Manipulation Data($\mathcal{D}$)
From Sensors
Observations
S_t
S_{t+1}
During Execution
State transitions
From End Effectors
Contact events
From Robotic Arms
Motion sequences
$\mathcal{D}_1$
Skill-level Datasets
$\mathcal{D}_2$
Task-level Datasets
$\mathcal{D}_3$
Foundation-level Datasets
Data Refinement
Visual and Viewpoint
Physics-Grounded
Base Policy
Motion & Force Corrections
Correction and Recovery
Curation and Verification
VR/AR
UMI
Handheld
Teleportation
Real-World Data Collection
Dataset
Benchmark
Data Acquisition
Automatic Data Generation
Ai2THOR
MuJoCo
CoppeliaSim
ManiSkill
NVIDIA
Simulation Data Ecosystems
Data Pipeline
GR00T-Teleop
Teleop Demonstration
Control Signals
Robot State
Isaac Lab
GR00T-Mimic
Motion Annotator
Isaac Lab
Trajectory Generation
Accelerated Physics Engine
Trajectory Evaluator
GROOT-Gen
Isaac Sim
3D
Cosmos